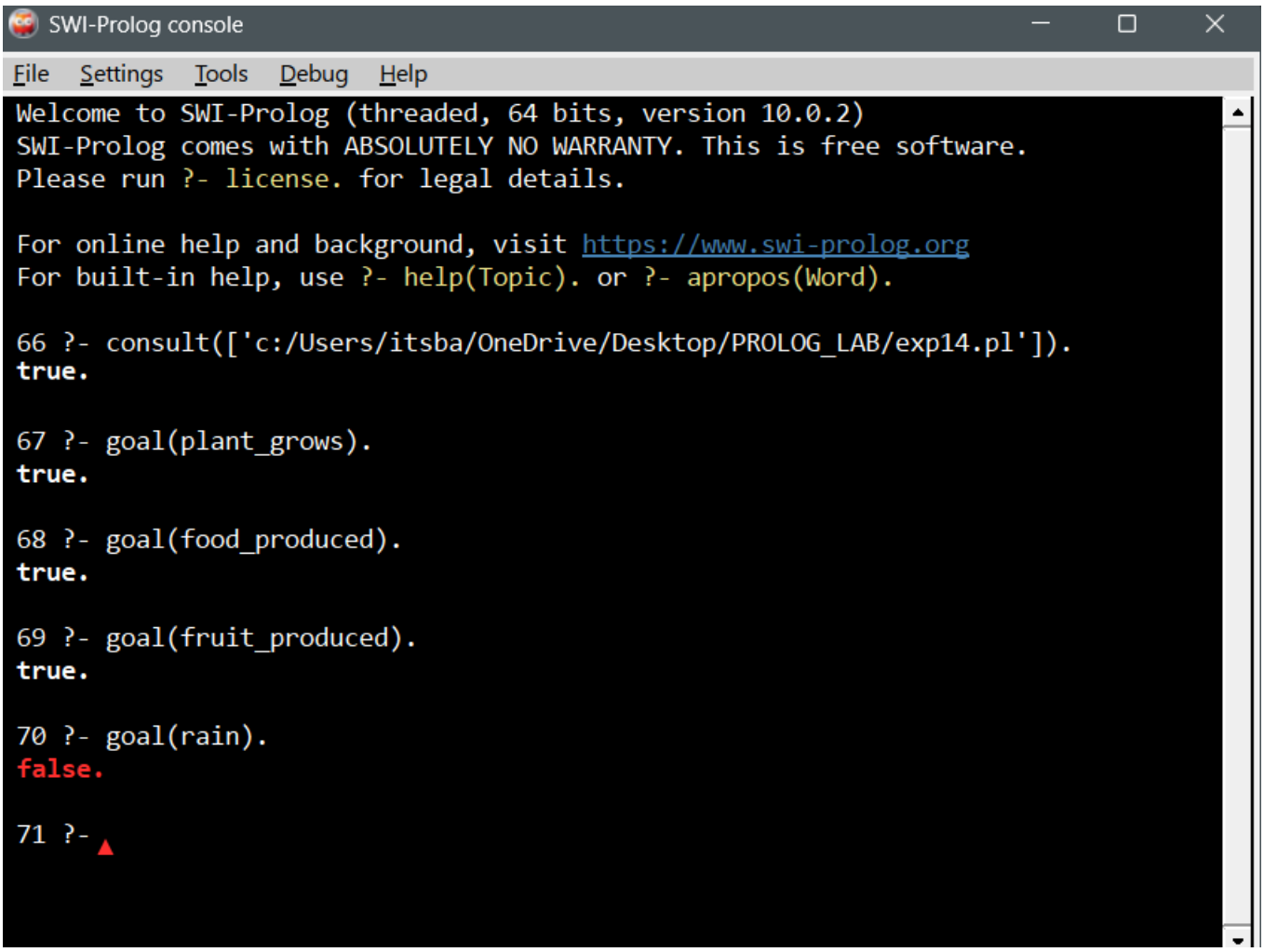
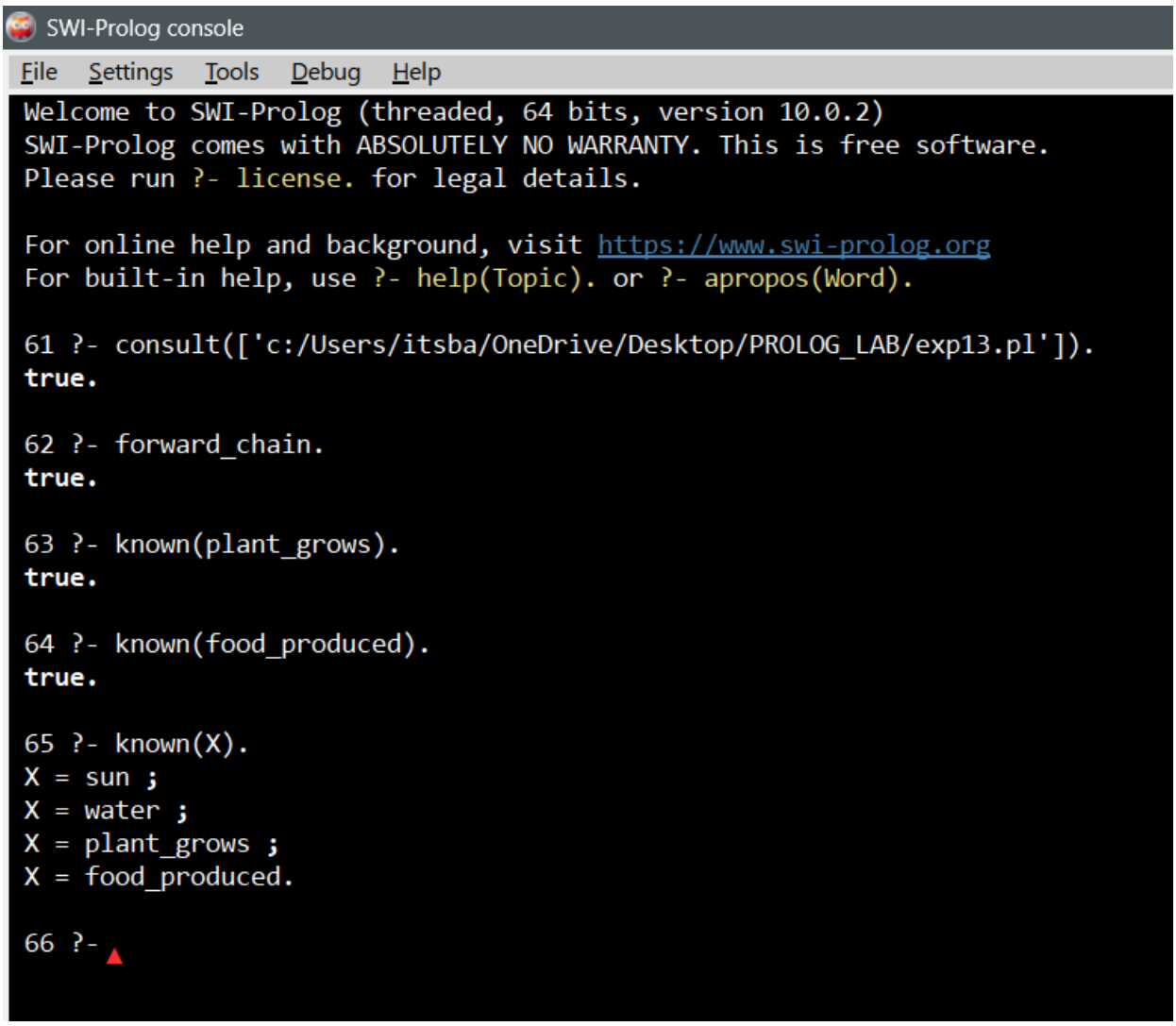
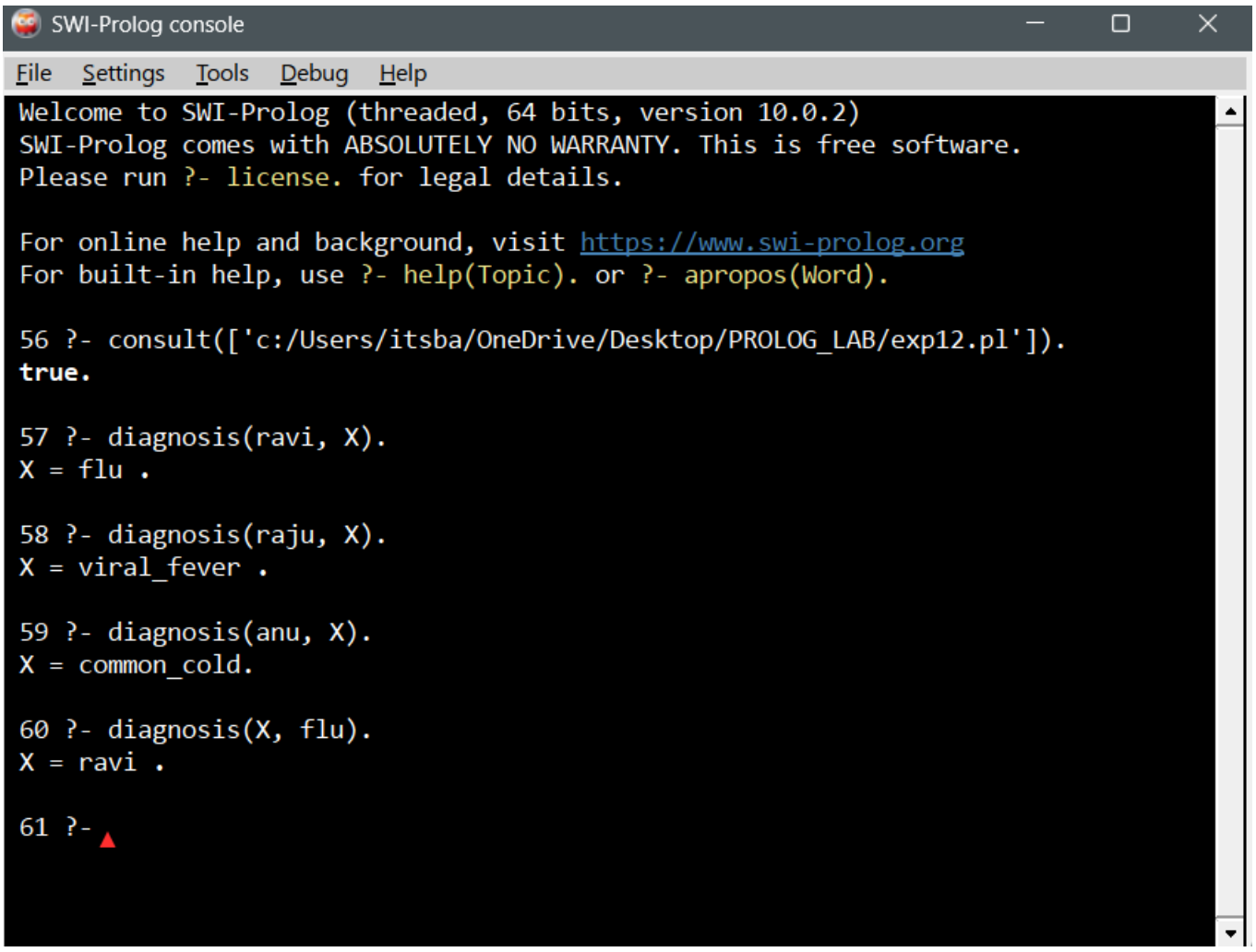
EXP 1


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For built-in help, use ?- help(Topic). or ?- apropos(Word).
66 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp14.pl']).
true.
67 ?- goal(plant_grows).
true.
68 ?- goal(food_produced).
true.
69 ?- goal(fruit_produced).
true.
70 ?- goal(rain).
false.
71 ?-

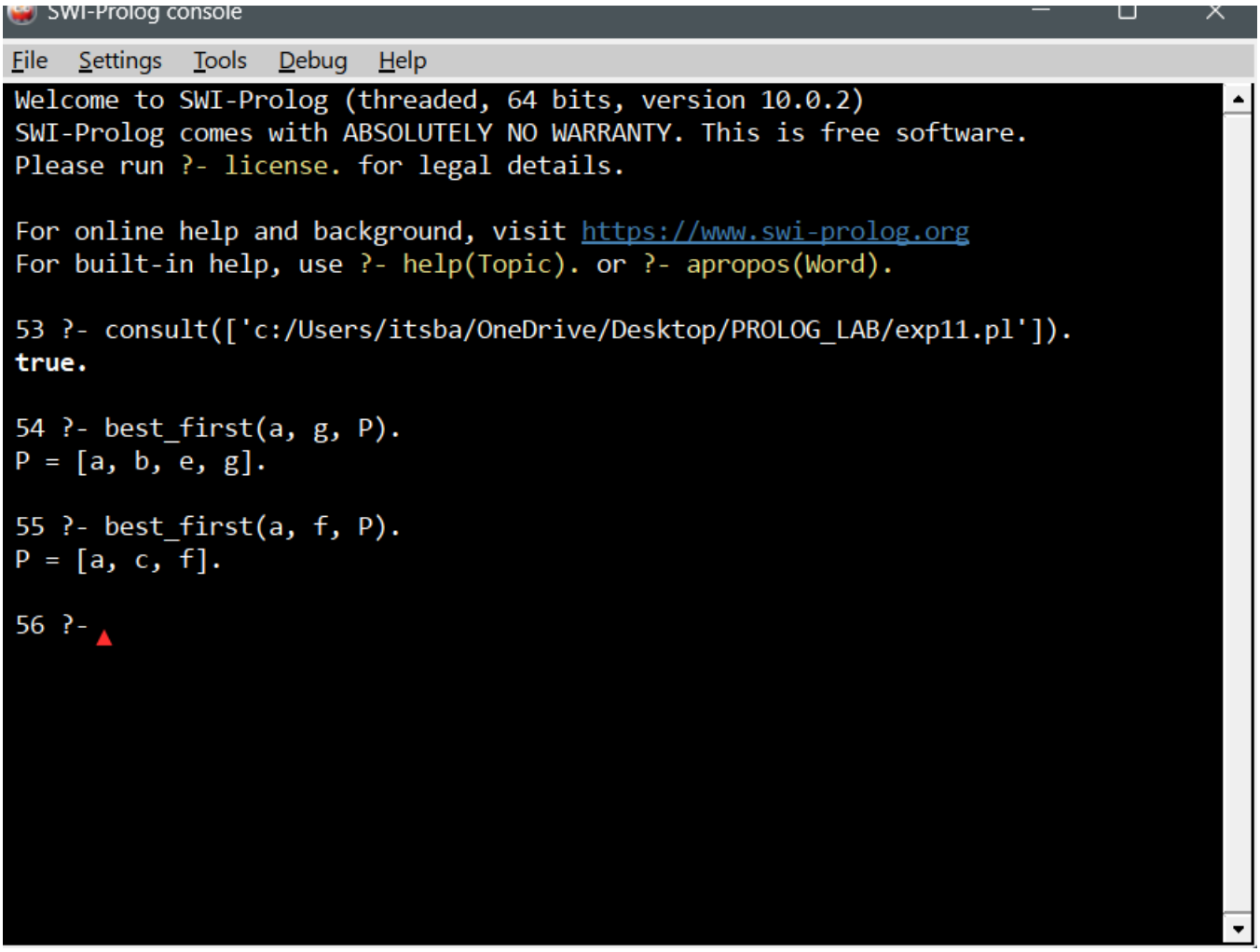
EXP 2


SWI-Prolog console
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For built-in help, use ?- help(Topic). or ?- apropos(Word).
61 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp13.pl']).
true.
62 ?- forward_chain.
true.
63 ?- known(plant_grows).
true.
64 ?- known(food_produced).
true.
65 ?- known(X).
X = sun ;
X = water ;
X = plant_grows ;
X = food_produced.
66 ?-

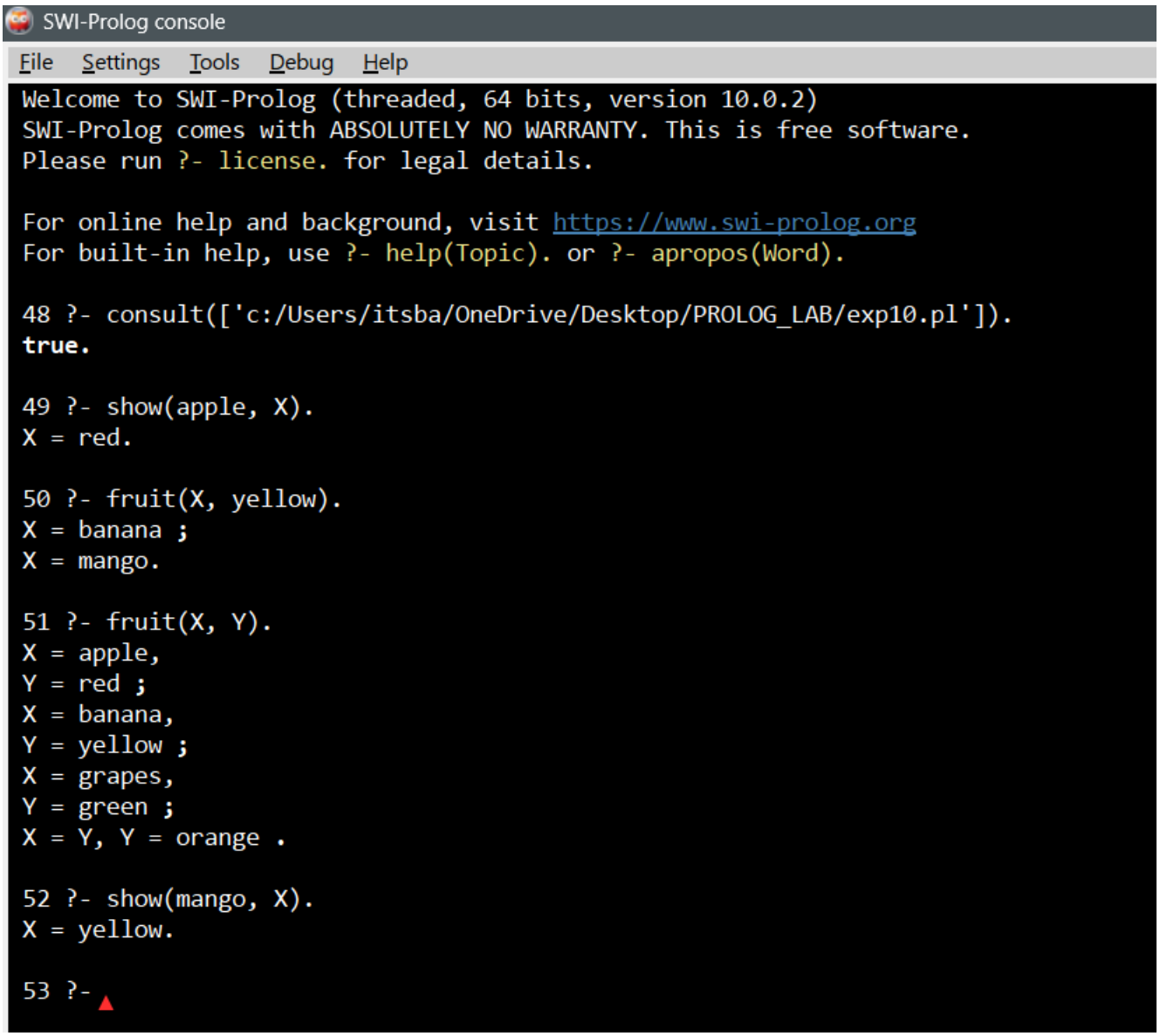
EXP 3


SWI-Prolog console
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For built-in help, use ?- help(Topic). or ?- apropos(Word).
56 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp12.pl']).
true.
57 ?- diagnosis(ravi, X).
X = flu .
58 ?- diagnosis(raju, X).
X = viral_fever .
59 ?- diagnosis(anu, X).
X = common_cold.
60 ?- diagnosis(X, flu).
X = ravi .
61 ?-

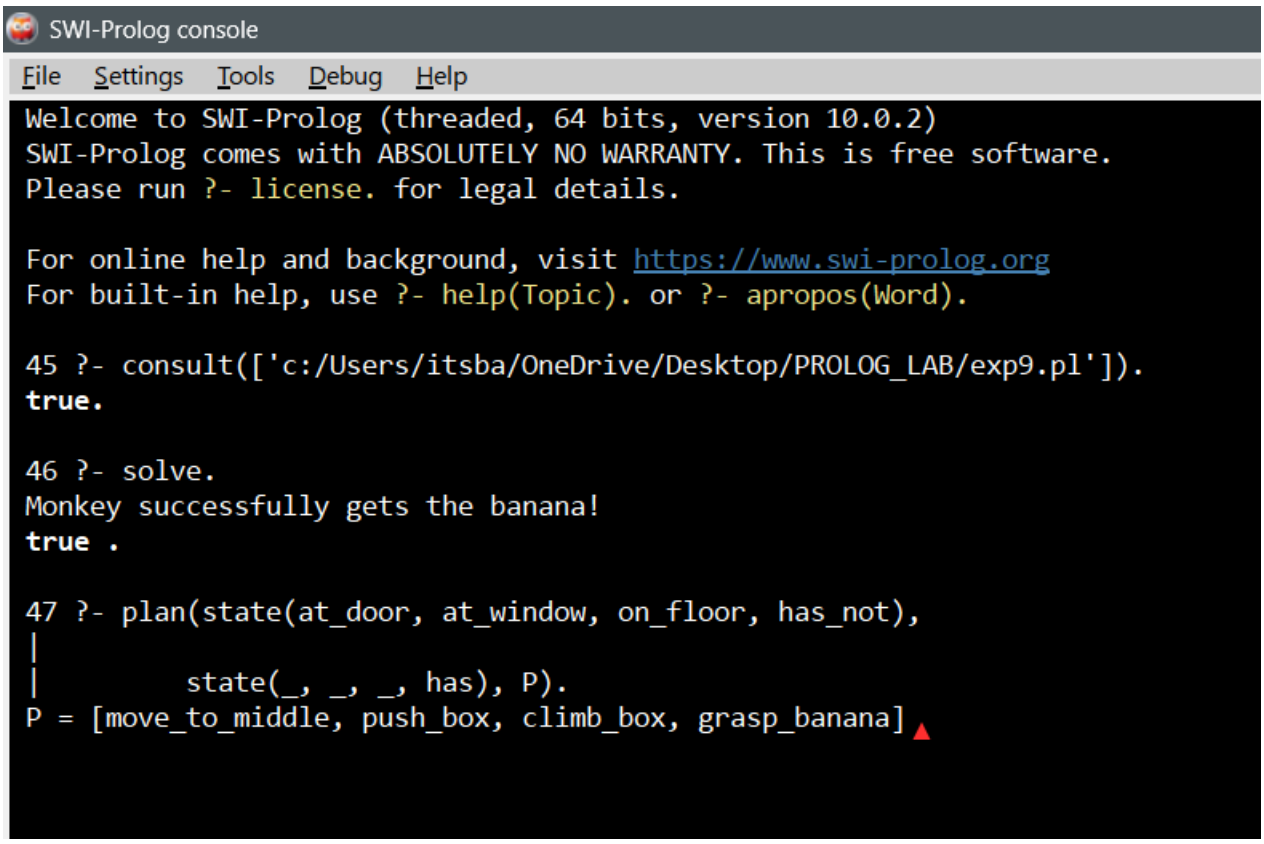
EXP 4


SWI-Prolog console
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For built-in help, use ?- help(Topic). or ?- apropos(Word).
53 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp11.pl']).
true.
54 ?- best_first(a, g, P).
P = [a, b, e, g].
55 ?- best_first(a, f, P).
P = [a, c, f].
56 ?-

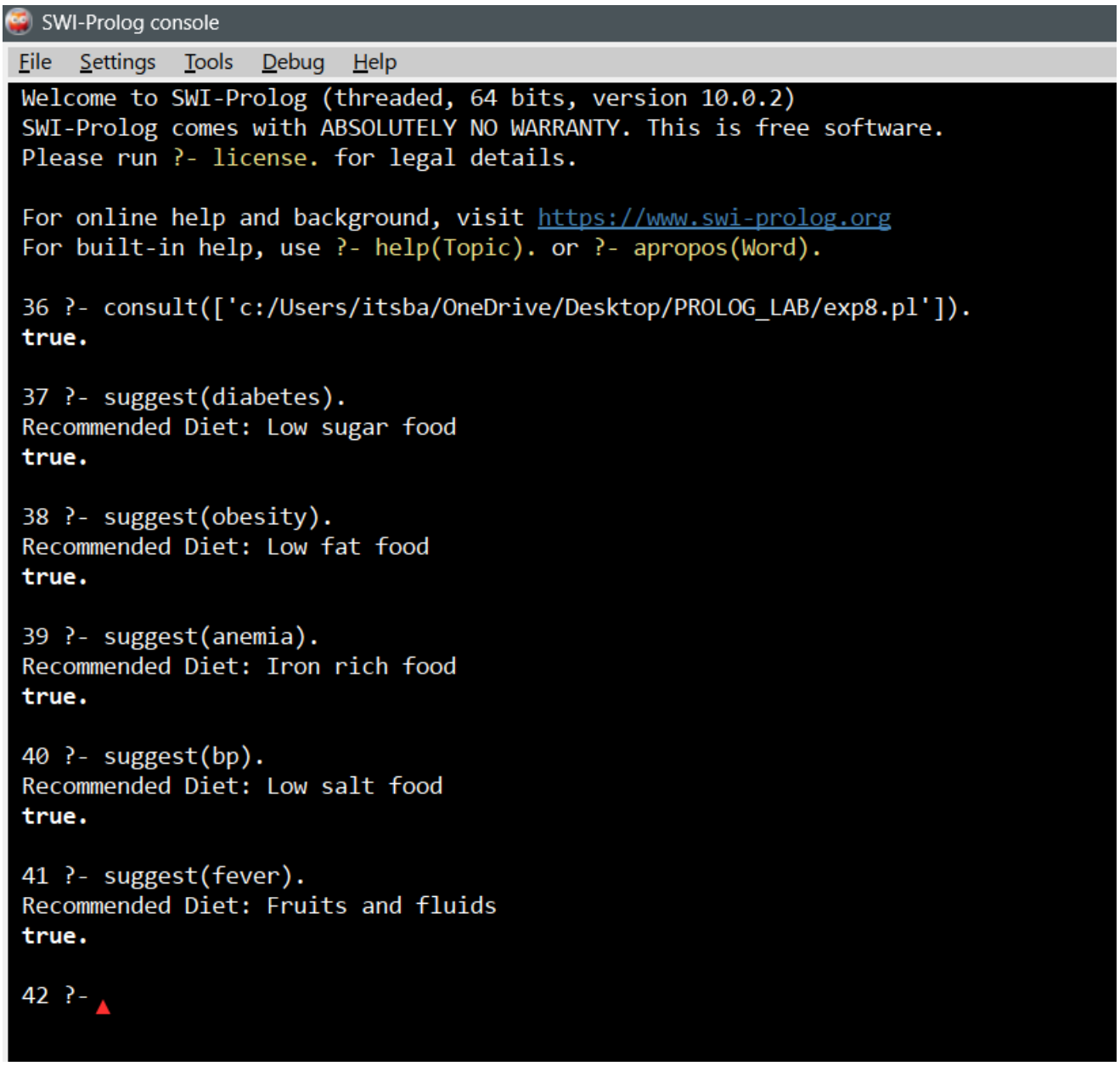
EXP 5


SWI-Prolog console
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
48 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp10.pl']).
true.
49 ?- show(apple, X).
X = red.
50 ?- fruit(X, yellow).
X = banana ;
X = mango.
51 ?- fruit(X, Y).
X = apple,
Y = red ;
X = banana,
Y = yellow ;
X = grapes,
Y = green ;
X = Y, Y = orange .
52 ?- show(mango, X).
X = yellow.
53 ?-

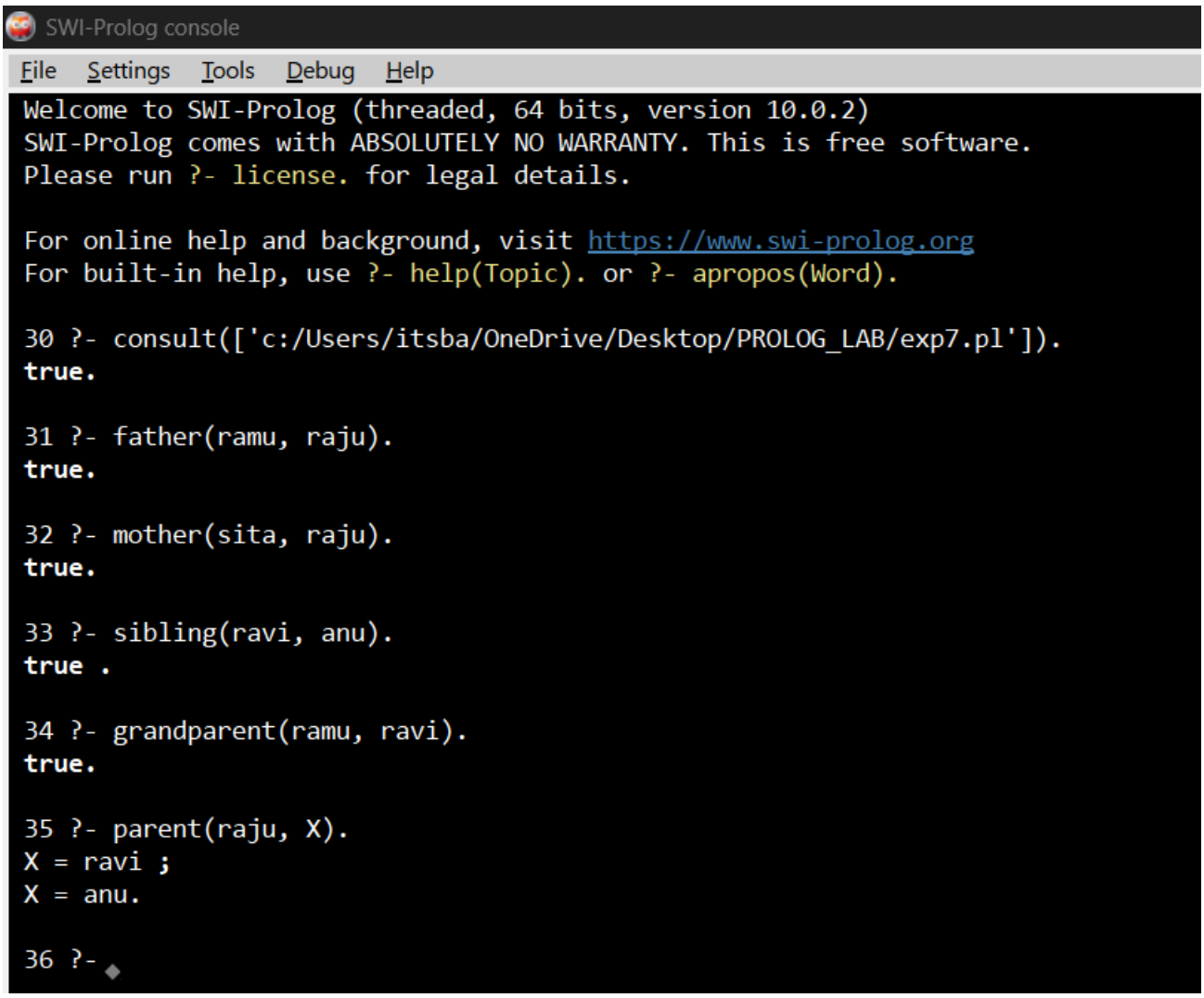
EXP 6


SWI-Prolog console
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
45 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp9.pl']).
true.
46 ?- solve.
Monkey successfully gets the banana!
true .
47 ?- plan(state(at_door, at_window, on_floor, has_not),
|
| state(_, _, _, has), P).
P = [move_to_middle, push_box, climb_box, grasp_banana]

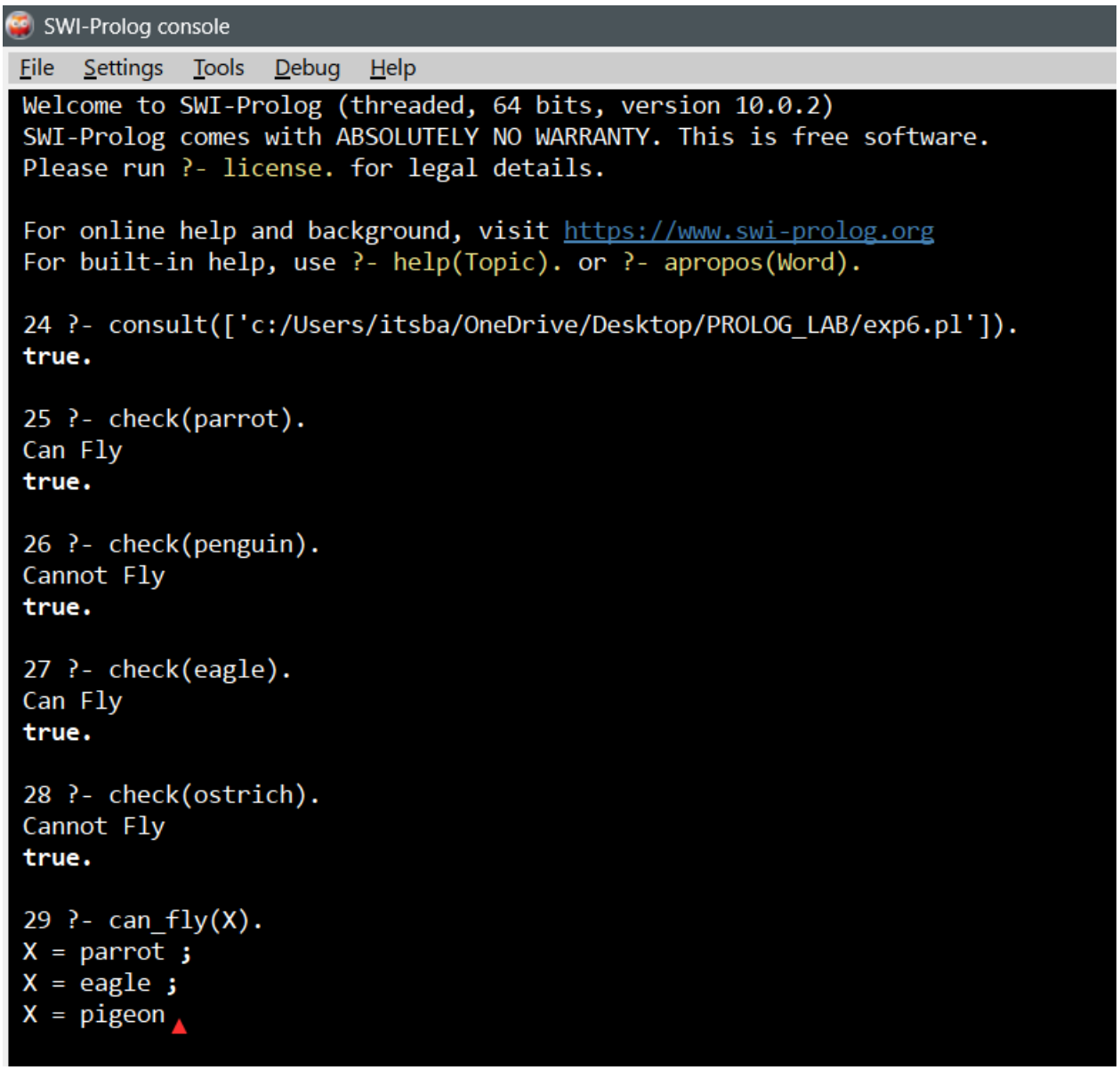
EXP 7


SWI-Prolog console
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
36 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp8.pl']).
true.
37 ?- suggest(diabetes).
Recommended Diet: Low sugar food
true.
38 ?- suggest(obesity).
Recommended Diet: Low fat food
true.
39 ?- suggest(anemia).
Recommended Diet: Iron rich food
true.
40 ?- suggest(bp).
Recommended Diet: Low salt food
true.
41 ?- suggest(fever).
Recommended Diet: Fruits and fluids
true.
42 ?-

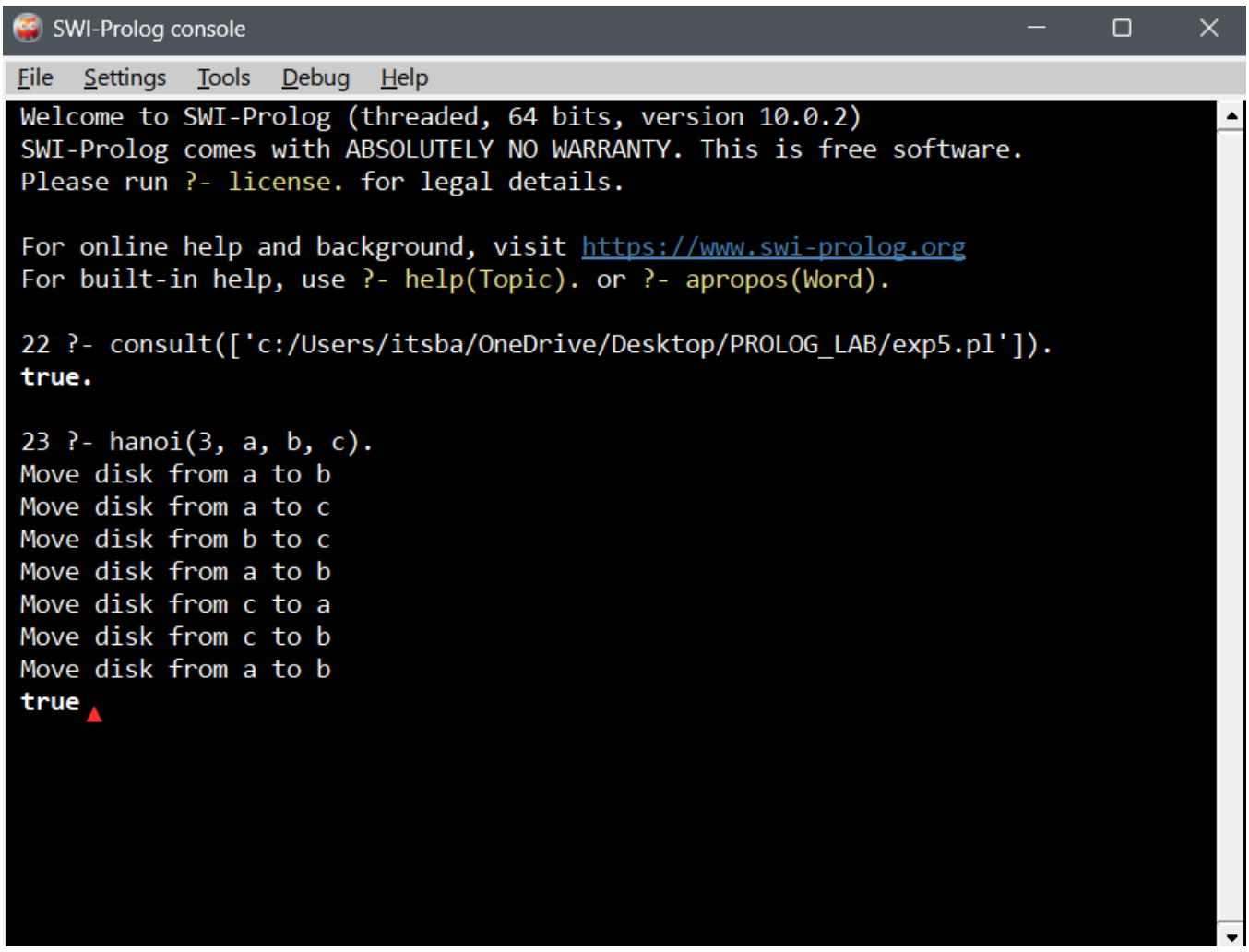
EXP 8


SWI-Prolog console
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
30 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp7.pl']).
true.
31 ?- father(ramu, raju).
true.
32 ?- mother(sita, raju).
true.
33 ?- sibling(ravi, anu).
true .
34 ?- grandparent(ramu, ravi).
true.
35 ?- parent(raju, X).
X = ravi ;
X = anu.
36 ?-

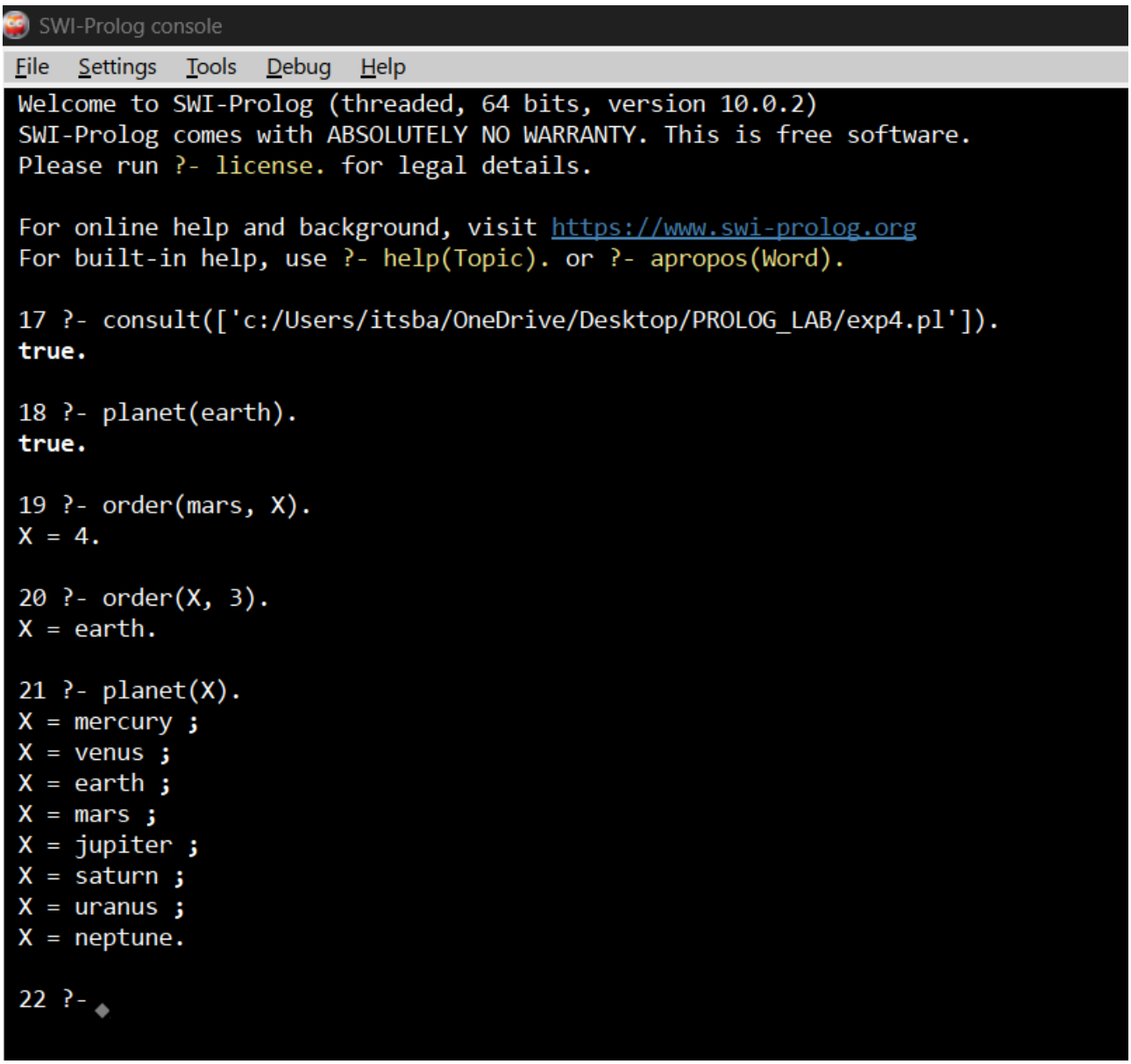
EXP 9


SWI-Prolog console
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For built-in help, use ?- help(Topic). or ?- apropos(Word).
24 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp6.pl']).
true.
25 ?- check(parrot).
Can Fly
true.
26 ?- check(penguin).
Cannot Fly
true.
27 ?- check(eagle).
Can Fly
true.
28 ?- check(ostrich).
Cannot Fly
true.
29 ?- can_fly(X).
X = parrot ;
X = eagle ;
X = pigeon

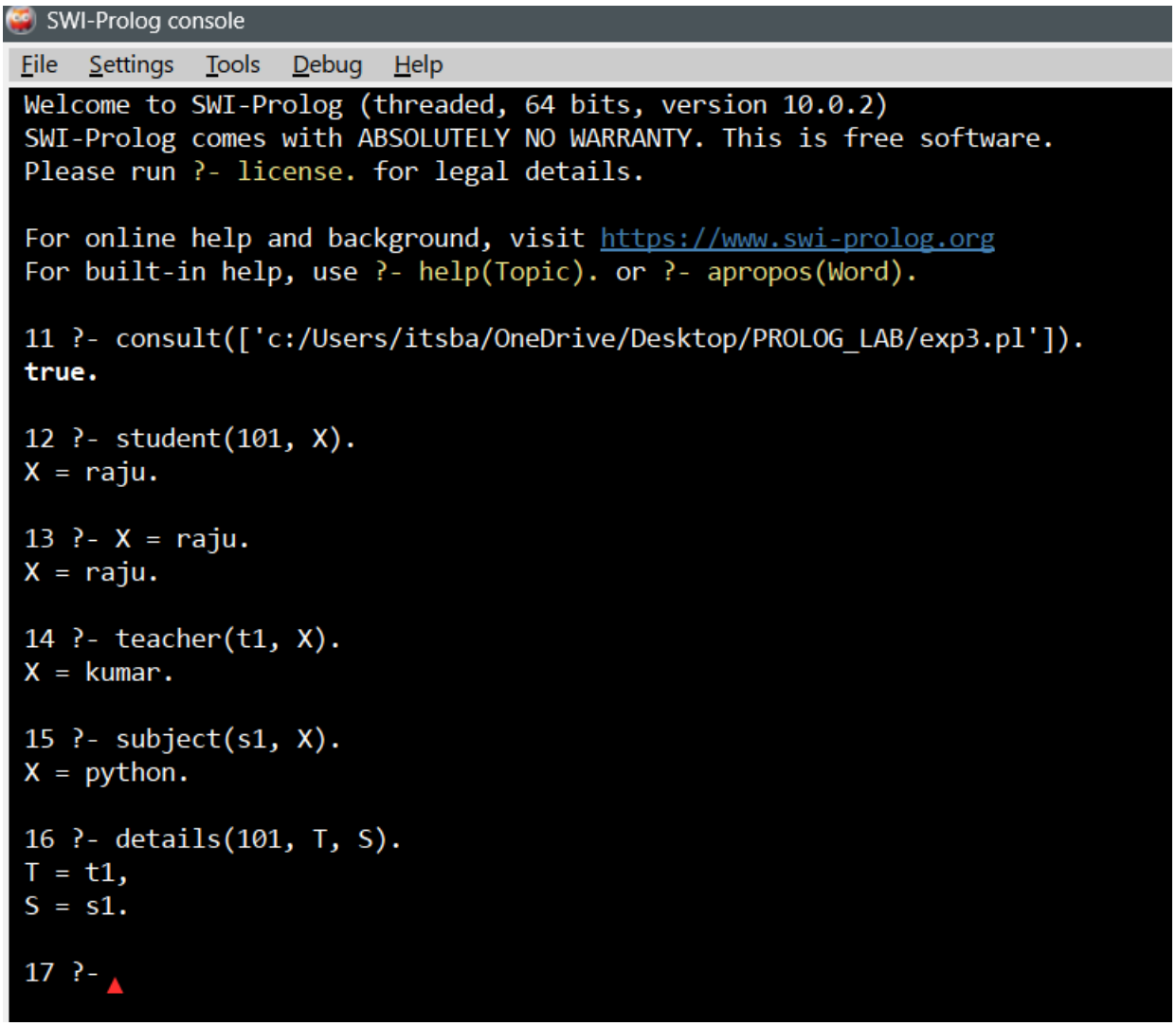
EXP 10


SWI-Prolog console
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
22 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp5.pl']).
true.
23 ?- hanoi(3, a, b, c).
Move disk from a to b
Move disk from a to c
Move disk from b to c
Move disk from a to b
Move disk from c to a
Move disk from c to b
Move disk from a to b
true

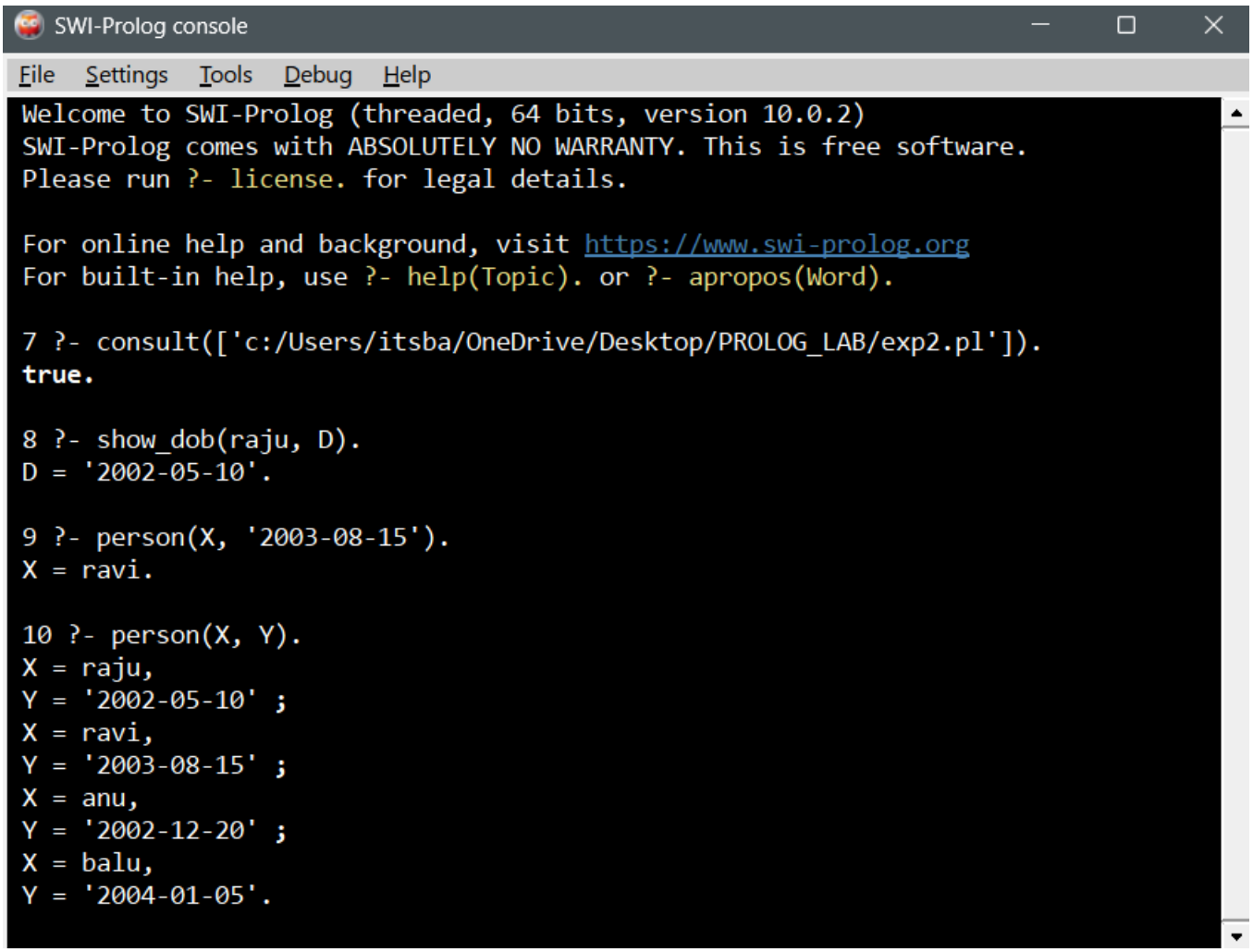
EXP 11


SWI-Prolog console
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
17 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp4.pl']).
true.
18 ?- planet(earth).
true.
19 ?- order(mars, X).
X = 4.
20 ?- order(X, 3).
X = earth.
21 ?- planet(X).
X = mercury ;
X = venus ;
X = earth ;
X = mars ;
X = jupiter ;
X = saturn ;
X = uranus ;
X = neptune.
22 ?-

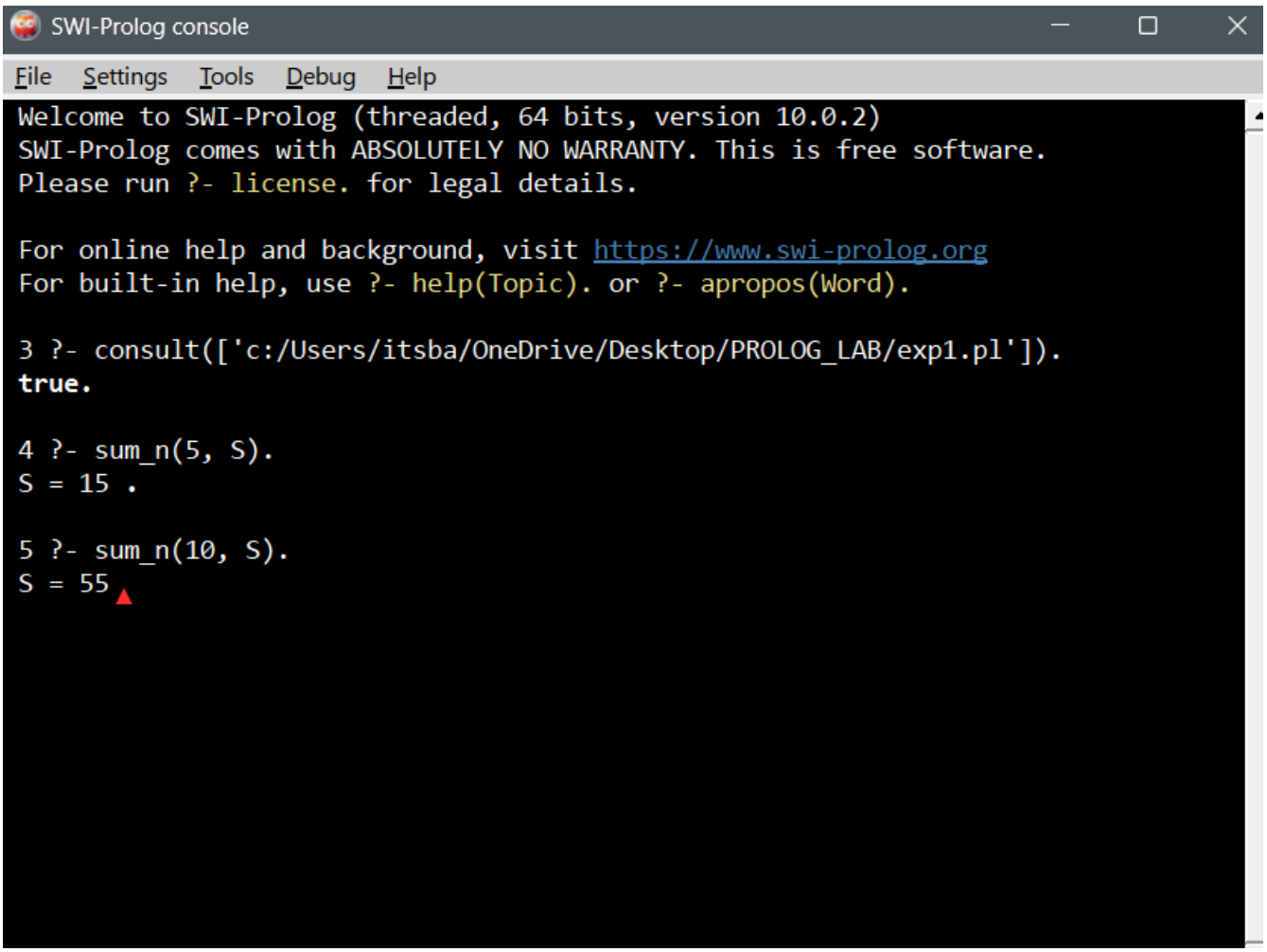
EXP 12


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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
11 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp3.pl']).
true.
12 ?- student(101, X).
X = raju.
13 ?- X = raju.
X = raju.
14 ?- teacher(t1, X).
X = kumar.
15 ?- subject(s1, X).
X = python.
16 ?- details(101, T, S).
T = t1,
S = s1.
17 ?-

EXP 13


SWI-Prolog console
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For built-in help, use ?- help(Topic). or ?- apropos(Word).
7 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp2.pl']).
true.
8 ?- show_dob(raju, D).
D = '2002-05-10'.
9 ?- person(X, '2003-08-15').
X = ravi.
10 ?- person(X, Y).
X = raju,
Y = '2002-05-10' ;
X = ravi,
Y = '2003-08-15' ;
X = anu,
Y = '2002-12-20' ;
X = balu,
Y = '2004-01-05'.

EXP 14


SWI-Prolog console
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For online help and background, visit https://www.swi-prolog.org
For built-in help, use ?- help(Topic). or ?- apropos(Word).
3 ?- consult(['c:/Users/itsba/OneDrive/Desktop/PROLOG_LAB/exp1.pl']).
true.
4 ?- sum_n(5, S).
S = 15 .
5 ?- sum_n(10, S).
S = 55